

Opeyemi Peter Ojajuni, PhD

Applied Scientist | Machine Learning, Human-Centered Evaluation & LLM Systems

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SUMMARY

Applied scientist (PhD) working across experimental design, machine learning and evaluation of AI and immersive systems. Evaluated a VR instruction intervention for 987 students over four years (ASEE 2026), and built a federated intrusion-detection pipeline reaching 0.92 malicious recall on a frozen 199K-sample test set while documenting its false-positive rate as the blocker to deployment. Builds working prototypes, including a citation-grounded RAG assistant and a human-in-the-loop LLM agent app. 20 publications, 125 citations. AWS Certified AI Practitioner.

TECHNICAL SKILLS

Machine learning: scikit-learn, Random Forest, SVM, XGBoost, KNN, CNN, federated learning (Flower, PyTorch)

GenAI & agents: RAG, LangChain, FAISS, OpenAI API, embeddings, local LLMs (Ollama), human-in-the-loop agent design

Experimentation & statistics: quasi-experimental design, factor analysis, chi-square, effect sizes, time series (ARIMA), forecasting

Engineering: Python, SQL, FastAPI, Streamlit, Next.js, Docker, Git/GitHub

XR & simulation: Unity, VR/CAVE, digital twins, LiDAR reality capture, GIS (OpenStreetMap, ArcGIS)

EXPERIENCE

Postdoctoral Research Manager, SU XR Jags Research Laboratory

Feb 2024 – Present

Southern University and A&M College, Baton Rouge, LA

- Evaluated a VR/CAVE-based engineering-instruction intervention against traditional instruction for 987 students over four years (Fall 2021 – Spring 2025) using a quasi-experimental, between-subjects design (ASEE 2026).
- Built a mobility-aware federated intrusion-detection pipeline for connected vehicles (VeReMi data), benchmarked against FedAvg, FedProx, FedNova and FedBuff; reached 0.92 malicious recall and ROC AUC 0.79 on a frozen 199,663-sample test set with ~53% estimated communication savings, and documented the high false-positive rate as the blocker to deployment.
- Evaluated a VR professional-development program (NTIA Connecting Minority Communities) for 88 teachers across two 2025 cohorts (pre/post surveys, 20-item self-efficacy battery, open responses; Kirkpatrick model); Fall cohort self-efficacy rose from 5.39 to 8.72 on a 1–10 scale (+62%), and 85% of Spring participants planned to implement VR that school year (self-reported, no control group).
- Co-developed a reality-capture → digital-twin → VR → 3D-print learning workflow; in a 68-student quasi-experiment, pass rates rose from 63.6% to 87.0% ($\chi^2(5) = 13.98$, $p = .016$) (journal manuscript, 2026).
- Built, trained and validated deep-learning (CNN) and classical ML models (SVM, XGBoost, Random Forest, KNN) on multimodal data, including CNN work for the Louisiana Department of Health ICLERT project.
- Built 10 immersive crisis-leadership training modules (2025) and trained 35+ crisis leaders.
- Co-PI on a 2026 research proposal for explainable, protocol-grounded LLM decision support in disaster preparedness, including a human-centred evaluation testbed built on immersive emergency simulations.

Data Scientist Intern, Amazon (Amazon Science – Mechatronics & Sustainable Packaging)

Jun 2023 – Sep 2023

Seattle, WA

- Ran the end-to-end analytics workflow for an automated-fulfillment initiative: problem framing, hypothesis design, modeling, validation, and leadership-facing analysis.
- Wrangled 1M+ logistics and transactional records with Python and SQL; built capacity-forecasting models with quantified uncertainty to inform capital-investment planning.
- Built data-transformation frameworks and simulation-ready analytical layers that made models more robust to operational variability and let product stakeholders self-serve the analysis.

Graduate Research Assistant, Southern University and A&M College

Aug 2018 – Dec 2023

Baton Rouge, LA

- Validated a 29-item, 5-factor computational-thinking scale with engineering students using factor analysis (ASEE 2024), then measured medium effect sizes for algorithmic thinking and problem-solving under CAVE instruction (ASEE 2025).
- Built a Random Forest student-success classifier on the UCI Student Performance data and shipped it as a public Streamlit web app (SITE 2023).
- Designed a software-defined-networking framework that detects and mitigates DDoS attacks on IoT networks through API-based bandwidth policies (IJTD, 2020).

Earlier: Network & Security Engineer, Staunton and Lycett Ltd, Nigeria (Mar 2015 – Feb 2018)

SELECTED PROJECTS

- Built InsightBot, a retrieval-augmented research assistant (LangChain, OpenAI embeddings, FAISS, Streamlit) that ingests article URLs and answers with source citations; public demo.
- Building Pathloom, a local-first career-intelligence app (Next.js, FastAPI, Ollama) in which an LLM agent proposes where documents belong and a human approves every change to stored facts; 33 automated test files.
- Designed a reproducible, leakage-safe data pipeline for vehicle-message anomaly analysis: 60,785 messages from VeReMi NextGen, official splits preserved, checksum-verified downloads, locked environment and tests.
- Built an airline revenue-management model (booking curves, demand forecasting, overbooking policy) showing ~10% revenue lift in simulation on synthetic data.

SELECTED PUBLICATIONS (20 PUBLICATIONS, 125 CITATIONS; FIRST-AUTHOR ML PAPER CITED 92 TIMES)

Ojajuni, O. P., Lawson, A. H. (2026). NSF IUSE: Evaluating academic performance in VR-enhanced engineering education: a quasi-experimental study. ASEE 2026.

Ojajuni, O., Ayeni, F., Akodu, O., et al. (2021). Predicting student academic performance using machine learning. ICCSA 2021, 481–491.

Ojajuni, O., Ismail, Y., Lawson, A. (2020). Distributed denial-of-service attack detection and mitigation for the Internet of Things. International Journal of Technology Diffusion, 11(2), 18–32.

EDUCATION

PhD, Science & Mathematics Education, Southern University and A&M College, 2023. Dissertation: Exploring the Impact of Immersive Technology on Engineering Students' Computational Thinking Skills

MBA (in progress), Southern University and A&M College

MSc Mobile & Satellite Communication, University of Surrey, UK, 2014 · BEng Computer Engineering, Covenant University, Nigeria, 2012

CERTIFICATIONS

AWS Certified AI Practitioner (active through Jul 2028) · IBM Artificial Intelligence Practitioner (2022)